MUAZZAM BADSHAH KHAN

AI/ML Engineer

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PROFESSIONAL SUMMARY

Motivated AI/ML Engineer with 1+ years of hands-on experience building machine learning and deep learning solutions through internships, personal, and client projects. Skilled in developing computer vision, medical image analysis, cybersecurity, and natural language processing applications using Python, PyTorch, and TensorFlow. Proficient in the full ML pipeline from dataset creation and preprocessing to model training, evaluation, and deployment. Passionate about applying AI to solve real-world problems and seeking an AI/ML internship to contribute to impactful projects while continuing to grow technically.

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript, C#, SQL
- AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
- Libraries: NumPy, Pandas, OpenCV, Librosa, Matplotlib, Seaborn, NLTK
- AI Concepts: Deep Learning, Convolutional Neural Networks (CNN), Computer Vision, Natural Language Processing (NLP), LangChain, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs)
- Data Management: Dataset creation, data preprocessing, data augmentation, feature engineering, data manipulation
- Tools & Platforms: VS Code, Jupyter Notebook, Google Colab, Kaggle, Git, GitHub, Docker
- Web Development: MERN Stack (MongoDB, Express.js, React, Node.js), HTML, CSS, .NET Core
- Other: Cryptography, Cybersecurity Fundamentals, Model Deployment, API Integration

EXPERIENCE

AI/ML Intern — National Center of Artificial Intelligence (NCAI), NUST *Jul 2026 – Present*

Islamabad, Pakistan

- Contribute to applied AI/ML research and development projects at NCAI, one of Pakistan's leading centers for artificial intelligence research.
- Design, train, and evaluate deep learning models using PyTorch and TensorFlow for computer vision and related applications.
- Assist with dataset preparation, experimentation, and performance benchmarking under the guidance of the technical team.
- Document research findings and contribute to internal knowledge-sharing within the AI/ML team.

Data Science Intern — 10Pearls (SHINE Internship Program) *Jun 2026 – Sep 2026*

Islamabad, Pakistan

- Selected for the 10Pearls SHINE Internship Program, applying data science and machine learning techniques to real-world projects.
- Collected, cleaned, and preprocessed structured and unstructured datasets to prepare them for analysis and model training.
- Built and evaluated machine learning models using Python (Pandas, NumPy, Scikit-learn) to extract insights and support data-driven decisions.
- Created data visualizations and reports to communicate findings to technical and non-technical stakeholders.
- Collaborated with cross-functional teams in an agile environment to deliver project milestones on schedule.

PROJECTS

Kidney Disease Detection and Classification System

- Built a deep learning model to detect and classify kidney diseases including kidney stones and inflammation from ultrasound images.
- Designed and trained a Convolutional Neural Network (CNN) using PyTorch and TensorFlow for multi-class medical image classification.
- Performed dataset preparation, image preprocessing, and data augmentation to improve model generalization and accuracy.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix to ensure clinical reliability.

Ultrasound Image Classification for Disease Detection

- Developed a computer vision pipeline to classify ultrasound images and detect medical abnormalities.
- Applied transfer learning and CNN architectures to extract features and improve diagnostic accuracy.
- Implemented an end-to-end workflow including data labeling, model training, validation, and inference.

Malware Detection Using CNN

- Developed a malware detection system leveraging Convolutional Neural Networks to classify malicious vs. benign files.
- Converted binary files into image representations for visual pattern recognition by the CNN model.
- Achieved high detection accuracy through hyperparameter tuning and data preprocessing techniques.

AI-Powered File Encryption and Decryption Tool

- Built a secure file encryption and decryption system combining cryptographic algorithms with AI-based key management.
- Integrated machine learning techniques to enhance security and detect anomalies in file access patterns.
- Implemented using Python with cryptography libraries to ensure data confidentiality and integrity.

Personal AI Chatbot

- Developed a personal-use chatbot leveraging Large Language Models (LLMs) and the LangChain framework.
- Implemented Retrieval-Augmented Generation (RAG) to provide context-aware and accurate responses.
- Designed conversational flows and integrated APIs for natural and dynamic user interactions.

EDUCATION

Bachelor of Computer Science (BCS) *2021 – 2025*

- Hazara University, Mansehra | CGPA: 3.5 / 4.0

FSc (Pre-Engineering / Intermediate) *2018 – 2020*

- Al-Syed Garden Public School & College

Matriculation (SSC)

- Al-Syed Garden Public School & College

CERTIFICATIONS

- AI/ML Certification (NAVTTC – Prime Minister's Youth Skills Development Program, Quaid-i-Azam University) — Grade A+
- Microsoft Certified: Azure AI Fundamentals
- Website Security and Exploitation
- Python for Everyone
- MERN Stack Development
- Full Stack Web Development
- .NET Core Development

LANGUAGES

Pashto (Native) | Urdu (Fluent) | English (Conversational)